

Scalable Cloud Architecture

Tasks-1. Pick any one app you use daily (like Zomato, BookMyShow, or Spotify) and write a short description of a new feature you would want to build for it that would require cloud hosting and scaling.

App : BookMyShow

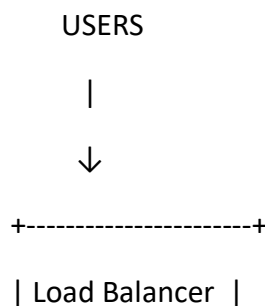
New feature: Personalized Event Recommendations

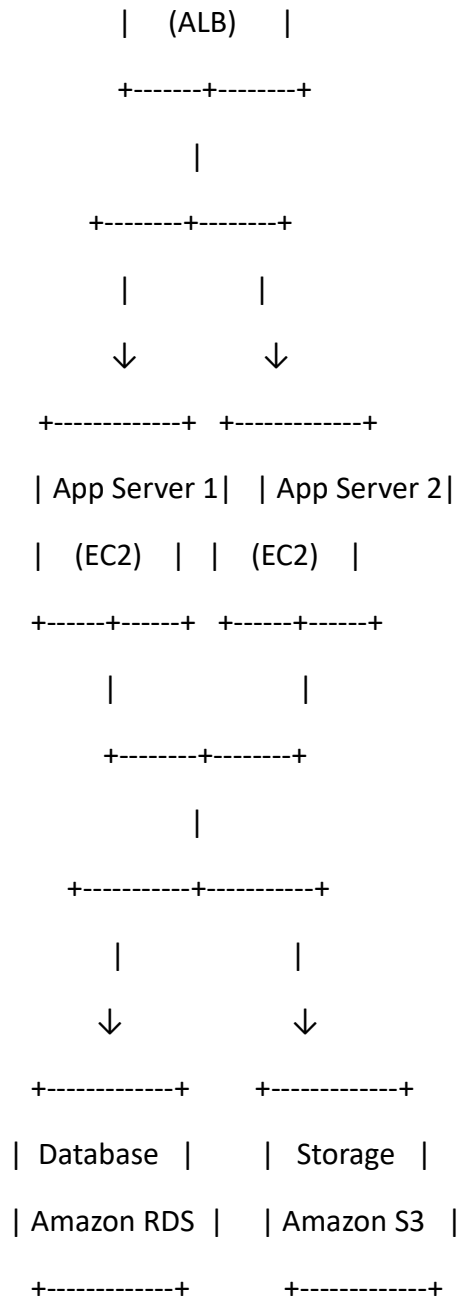
I would like to build a feature that recommends movies, concerts, sports matches, and other events based on a user's interests, previous bookings, and browsing history. The feature would require cloud hosting because it needs to process large amounts of user data and handle many users at the same time. It should also be able to scale during events such as IPL matches, popular movie releases, and concerts.

Tasks-2. Draw a basic cloud architecture diagram (hand-drawn or using a tool like draw.io) for your chosen feature, including at least: a load balancer, two app servers, a database, and a storage service (like AWS S3 or Azure Blob). Label each component clearly.

The architecture can use AWS services:

- Application Load Balancer (ALB) – distributes incoming requests.
- Two EC2 App Servers – process application requests.
- Amazon RDS – stores user and booking information.
- Amazon S3 – stores images, posters, and other files.
- Auto Scaling Group – adds/removes app servers based on demand.
- CloudWatch – monitors the application.





Auto Scaling Group

monitors/scales EC2

Tasks-3.List three security measures you would add to your architecture to protect user data and prevent attacks. For each, explain in 1-2 lines how it helps (e.g., HTTPS, firewalls, authentication methods).

1. HTTPS / SSL

HTTPS encrypts communication between users and the BookMyShow application. This prevents attackers from easily reading sensitive information such as login details and personal data.

2. Web Application Firewall (WAF)

AWS WAF can help block malicious web traffic, including common attacks such as SQL injection and unwanted automated requests, before they reach the application servers.

3. Authentication and IAM

Secure authentication such as strong passwords and MFA can protect user accounts. AWS IAM can ensure that cloud administrators and services receive only the permissions they actually need.

Tasks-4. Write a short document (200-300 words) explaining how your architecture would handle a sudden spike in users during a big event (like IPL finals for BookMyShow or a major album release on Spotify). Describe which parts would scale and how.

During a major event such as an IPL final, the number of users accessing BookMyShow can increase dramatically within a few minutes. The proposed architecture handles this traffic using a load balancer and auto-scaling. First, the Application Load Balancer (ALB) receives requests from users and distributes them across the available application servers. This prevents one server from becoming overloaded while another server remains unused.

The Auto Scaling Group monitors the application servers and automatically launches additional EC2 instances when demand increases. For example, if two servers are unable to handle the increased traffic, the system can add a third server and distribute requests across all available servers. When the event ends and traffic decreases, Auto Scaling can remove unnecessary servers and return to the minimum number of instances. This saves money because we do not need to keep many servers running all the time.

The Amazon RDS database stores important information such as users, bookings, and event details. It can be configured with appropriate scaling and high-availability features to handle increased database activity. Amazon S3 stores posters, images, and other large files and can handle a large number of requests without requiring us to manually add storage servers.

CloudWatch monitors CPU usage, traffic, and other metrics and can generate alerts when problems occur. Overall, load balancing and auto-scaling allow the application to remain

available and responsive during sudden traffic spikes while reducing resources when demand becomes normal.

Tasks-5.Present your architecture design to a friend or family member, ask them for one piece of feedback or a question, and write down what they said along with your response.Focus on explaining your choices simply, as if you were talking to someone new to cloud computing.

Friend's question:

"Why did you choose an Application Load Balancer instead of a Network Load Balancer?"

My response:

"I chose an Application Load Balancer because BookMyShow is mainly a web and mobile application that uses HTTP and HTTPS. An ALB works at the application level and can distribute web requests between multiple servers. It also supports features such as path-based routing, which can be useful if different parts of the application need to be handled by different servers."